7 Rose and Emma each wear a device that records the number of steps they take in a day. All the results for a 7-day period are given in Fig. 7.

Day	1	2	3	4	5	6	7
Rose	10014	11262	10149	9361	9708	9921	10369
Emma	9204	9913	8741	10015	10261	7391	10856

Fig. 7

The 7-day mean is the mean number of steps taken in the last 7 days. The 7-day mean for Rose is 10112.

(i) Calculate the 7-day mean for Emma. **[1]**

At the end of day 8 a new 7-day mean is calculated by including the number of steps taken on day 8 and omitting the number of steps taken on day 1. On day 8 Rose takes 10259 steps.

(ii) Determine the number of steps Emma must take on day 8 so that her 7-day mean at the end of day 8 is the same as for Rose. **[4]**

In fact, over a long period of time, the mean of the number of steps per day that Emma takes is 10341 and the standard deviation is 948.

(iii) Determine whether the number of steps Emma needs to take on day 8 so that her 7-day mean is the same as that for Rose in part **(ii)** is unusually high. **[3]**

7(i)	
7(ii)	

7(iii)	